Enhancing Wellbeing in Jordan - Technical Report

Presented to the Data Visualization and Exploration Course,

Department of Computer Information Systems and Data Science,

Faculty of Computer and Information Technology,

Jordan University of Science and Technology

By:

Mohammad Abdulelah Faris

Supervised By:

Dr. Amera Al Amery

Irbid, Jordan

1st Semester 2025/2026

1. Abstract

A study has been completed on 1000+ participants Jordanian to evaluating and enhancing the subjective well-being of the population. It's about four subjects: (The Food Insecurity Experience Scale (FIES), Resilience Scale (RS), Mental disorder-Kessler Psychological Distress Scale (K10+), Stress Mindset(SMM)), the study had done by ministry of health in Jordan using subjective Wellbeing Questionnaire (SWQ) survey, this report will answer five questions that have been wrote to find some relation effects that may happen to us.

2. Introduction

This study helps the Ministry of Health in Jordan understand the importance of fostering positive experiences and environments among the population. To achieve this, a Subjective Wellbeing Questionnaire (SWQ) survey was conducted, covering four areas: the Food Insecurity Experience Scale (FIES), Resilience Scale (RS), Mental Disorder—Kessler Psychological Distress Scale (K10+), and Stress Mindset (SMM). Each area is measured using numeric or categorical scores.

FIES consists of eight questions (Yes = 1, No = 0, Sometimes Yes) assessing access to adequate food; higher scores indicate lower food security. RS includes six statements scored 1–5 (1 = Strongly Disagree, 5 = Strongly Agree), reflecting participants' ability to recover from stress. K10+ has ten questions on psychological distress using a five-point scale; lower scores indicate better mental health. Stress Mindset measures beliefs about stress through eight statements, showing whether individuals see stress as helpful or harmful.

The SWQ analysis aimed to answer five questions: The impact of physical activity on stress mindset and Resilience, Are stress mindset scores affected by gender, Are university students experience better wellbeing than non-students, The relationship between food security and students' GPA, and The effect of family income on food choices. This report shows the analysis methods and presents the results and insights of these analyses.

3. Methodology

This study employed a quantitative approach to analyze the Subjective Wellbeing Questionnaire (SWQ) dataset. All data manipulation, cleaning, and statistical analyses were conducted using the R Statistical Computing Environment (utilizing the *tidyverse*, *ggplot2*, and *stats* packages).

3.1. Data Pre-processing and Transformation

Prior to analysis, the raw dataset underwent a rigorous cleaning process to ensure data integrity:

- **Handling Missing Data:** The dataset was inspected for missing values. To minimize bias and preserve the distribution structure, were imputed using the most frequent data, while missing values in some numerical variables were replaced with the median, each according to the situation.
- **Variable Encoding:** Categorical variables representing ordinal data (specifically Cumulative GPA and Household Income) were explicitly encoded as ordered factors. This transformation was essential to enable polynomial contrast analysis (linear, quadratic, and cubic trends) rather than treating them as nominal categories.
- **Score Categorization:** To enhance the analytical depth of the study, a new variable was generated by transforming the four numerical scores into categorical intervals.
- **Score Calculation:** New aggregate variables were constructed based on the project dictionary:
    - **Kessler Psychological Distress Scale (K10):** Calculated by summing the scores of 10 items (range: 10–50).
    - **Brief Resilience Scale (RS):** Items 2, 4, and 6 were reverse coded prior to summation to ensure consistent directionality. The final score was the sum of all six items (range: 6–30).
    - **Food Insecurity (FIES):** Calculated as the row sum of binary responses (Yes=1, No=0) and (not sure=0.5) across eight food security questions.
    - **Stress Mindset (SMM):** Items 1, 3, 5, and 7 were reverse coded prior to summation to ensure consistent directionality. The final score was the sum of all eight items (range: 0–32).
    - **Dietary Indices:** Two custom indices were generated: a Healthy Eating Score (sum of fruits, vegetables, whole grains, dairy, beans, and fish consumption) and an Unhealthy Eating Score (sum of fast food and caffeine consumption).

Note: sum of new columns created like the factor of GPA contains some NA values due to a non-students situation in the "Are you in collage" column, However It was dealt with while ordering the factor while doing statistics.

3.2. Statistical Analysis

- **Normality Assessment:** The distribution of all outcome variables (K10, RS, FIES, SMM, Dietary Scores) was assessed using both visual inspection (histograms and Q-Q plots) and the Shapiro-Wilk test. Results indicated a significant deviation from normality ($p < .05$) for all key variables.
- **Group Comparisons:** Due to the non-normal distribution of the data, the Wilcoxon Rank-Sum Test and Linear Models were utilized to assess differences between groups.

  Also to investigate the relationship between ordered independent variables (GPA and Income) and continuous outcomes, Linear Model (LM) analyses were conducted. These models specifically tested for Linear Trends (monotonic increase/decrease) and higher-order polynomial trends (Quadratic/Cubic) to capture complex relationships.
- **Significance Level:** All statistical tests were two-tailed, and a significance level (alpha) of 0.05 was established as the threshold for statistical significance.

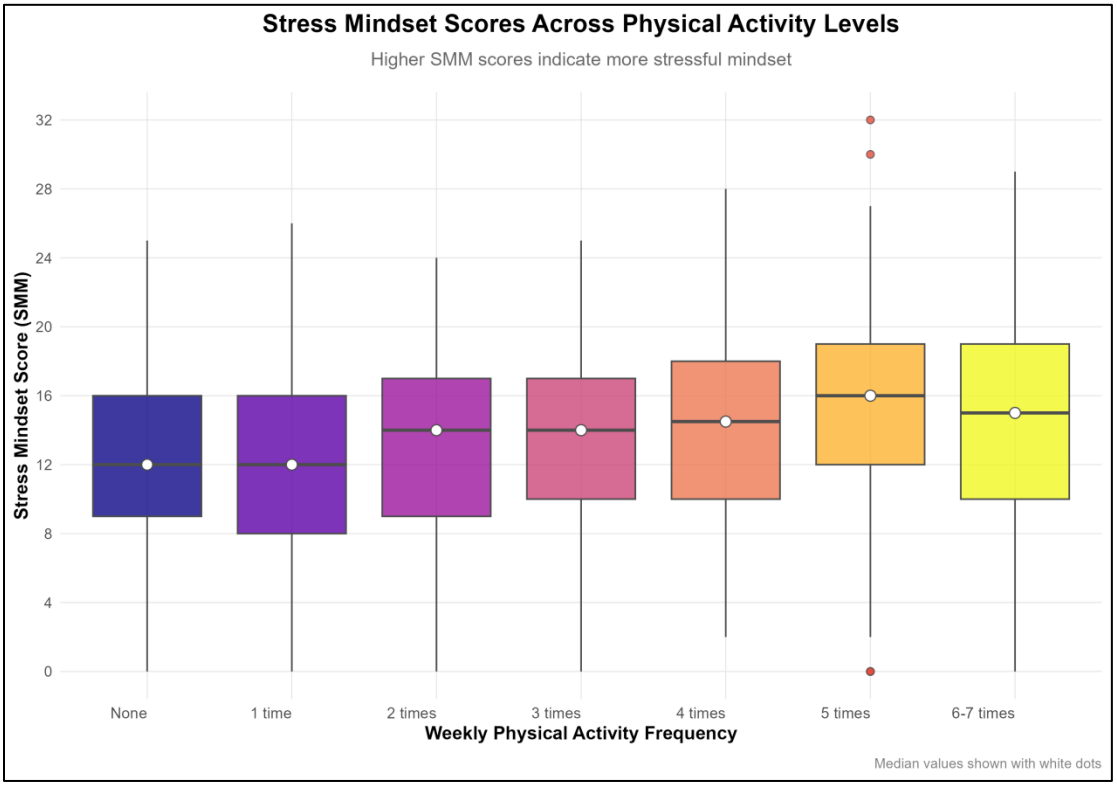
4. Results

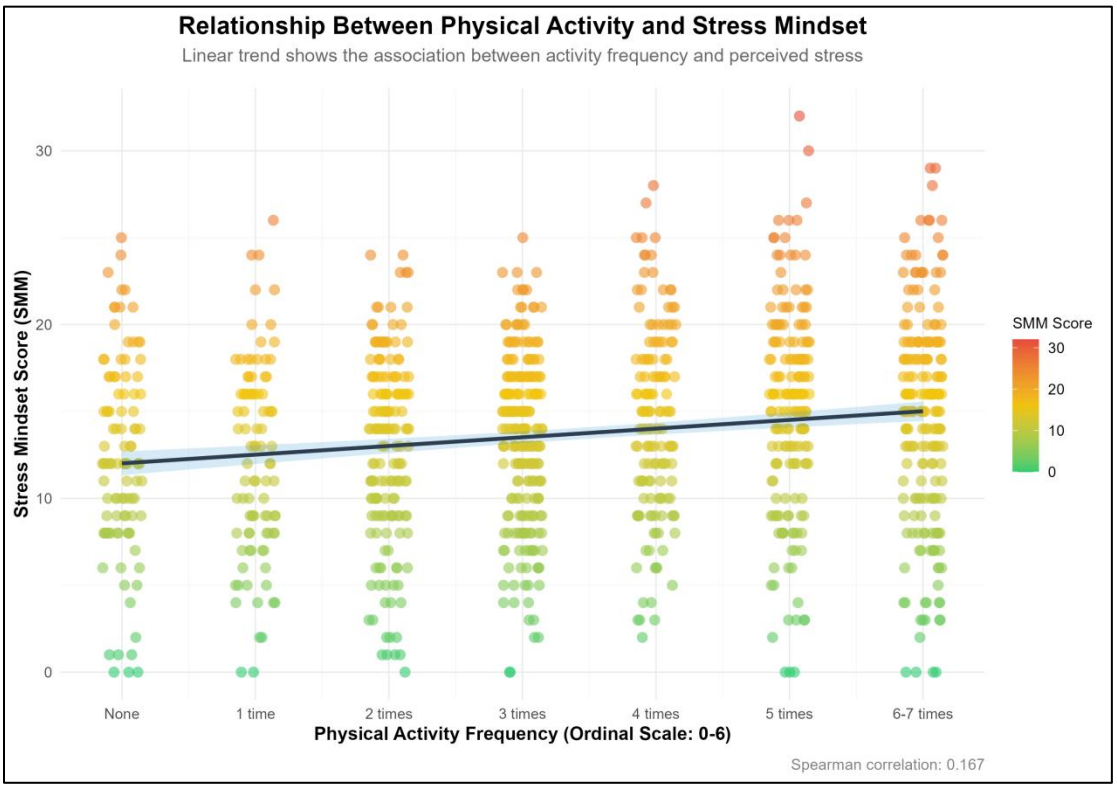
4.1 Q1: Do people with different levels of physical activity differ in their resilience (BRS) and stress mindset (SMM) scores?

To understand the relationship between physical activity and stress-related psychological factors, we build two linear models.

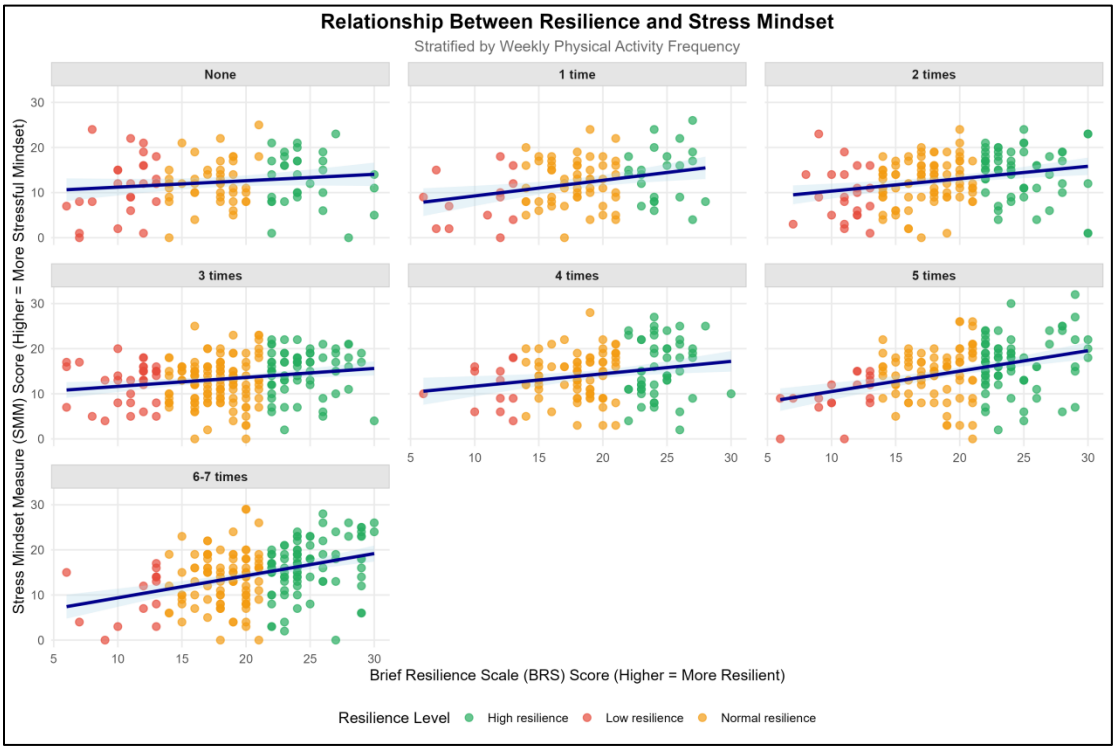
The first linear model showed a statistically significant association between SMM and physical activity. Higher SMM values were observed only among individuals engaging in physical activity four times per week or more, whereas lower activity frequencies were not significantly associated with SMM.

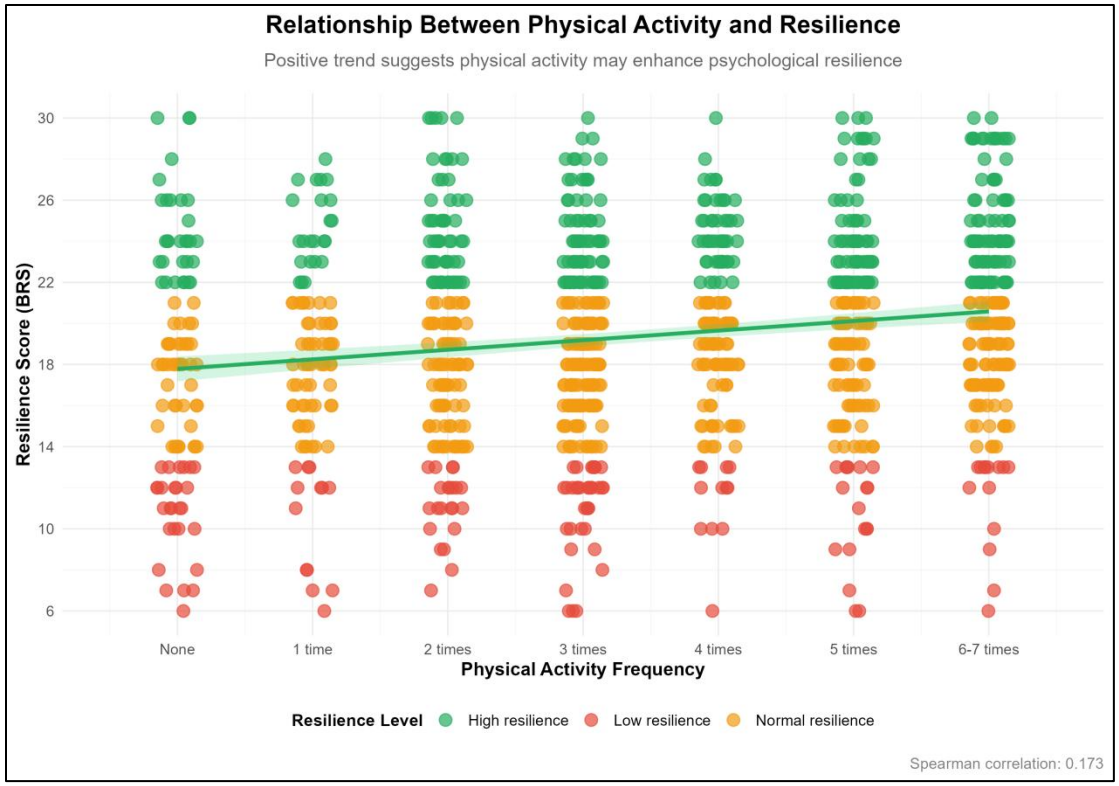
The second linear model showed a statistically significant association between BRS and physical activity frequency. Higher BRS values were observed among individuals engaging in physical activity two times per week or more, with stronger effects at four or more times per week, while lower activity frequencies were not significantly associated with BRS.


Stress Mindset Scores Across Physical Activity Levels
Higher SMM scores indicate more stressful mindset
Stress Mindset Score (SMM)
32
28
24
20
16
12
8
4
0
None
1 time
2 times
3 times
4 times
5 times
6-7 times
Weekly Physical Activity Frequency
Median values shown with white dots


Relationship Between Physical Activity and Stress Mindset
Linear trend shows the association between activity frequency and perceived stress
Stress Mindset Score (SMM)
30
20
10
0
None
1 time
2 times
3 times
4 times
5 times
6-7 times
Physical Activity Frequency (Ordinal Scale: 0-6)
SMM Score
30
20
10
0
Spearman correlation: 0.167

Another correlation test, it was between stress mindset and Resilience Score (BRS), sperman correlation was used after check the normality using Shapiro test, the result was (rho 0.29, p-value < 2.2e-16), There was a statistically significant, positive correlation between the two variables, tend to report higher resilience and recover from stress (More SMM -> slightly higher resilience), See the images that show us the relation between resilience, SMM, physical activity



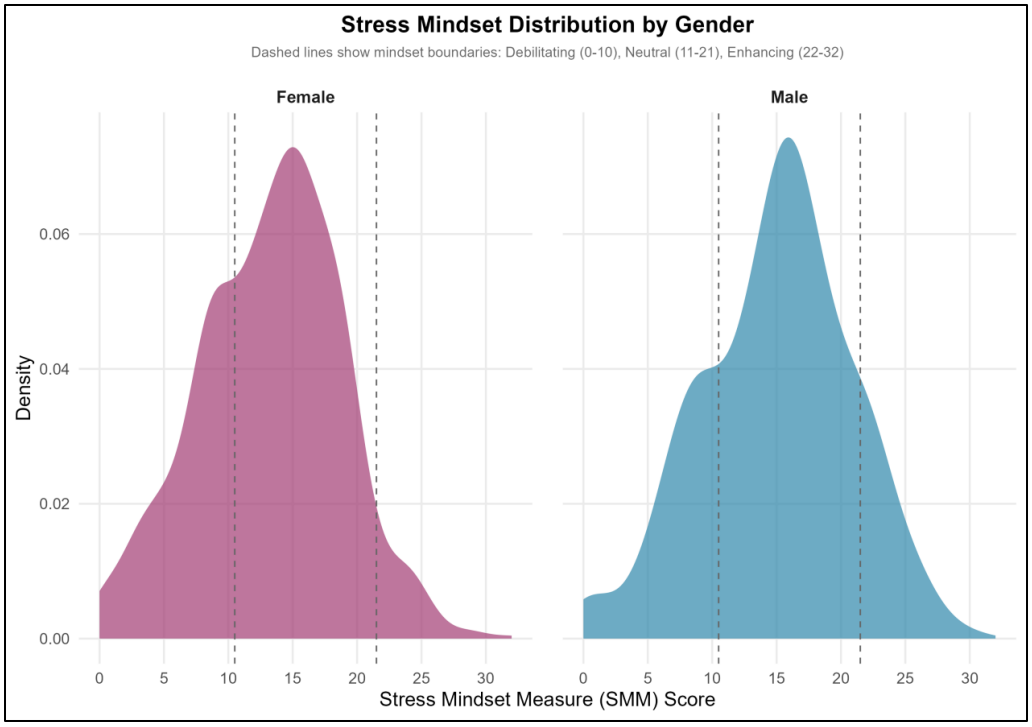


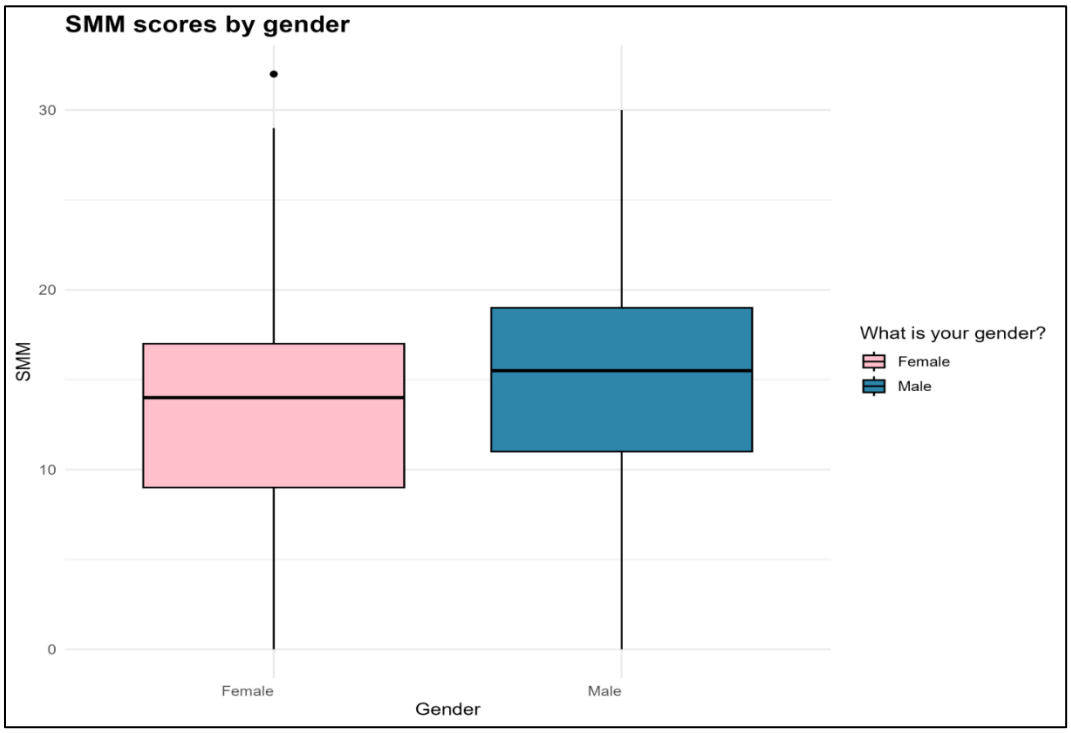
4.2 Q2: Does the stress mindset affected by gender?

To understand the relationship between the gender and the stress level, we apply the Wilcox test after check the normality using Shapiro test.

For Wilcox test, We do it between the gender column and stress mindset column, the result was (w = 97 , p-value = 1.041e-05). It was a statistically significant, this indicates that there is a difference between stress mindset and gender.

The visualization show us that males have a Higher stress than females

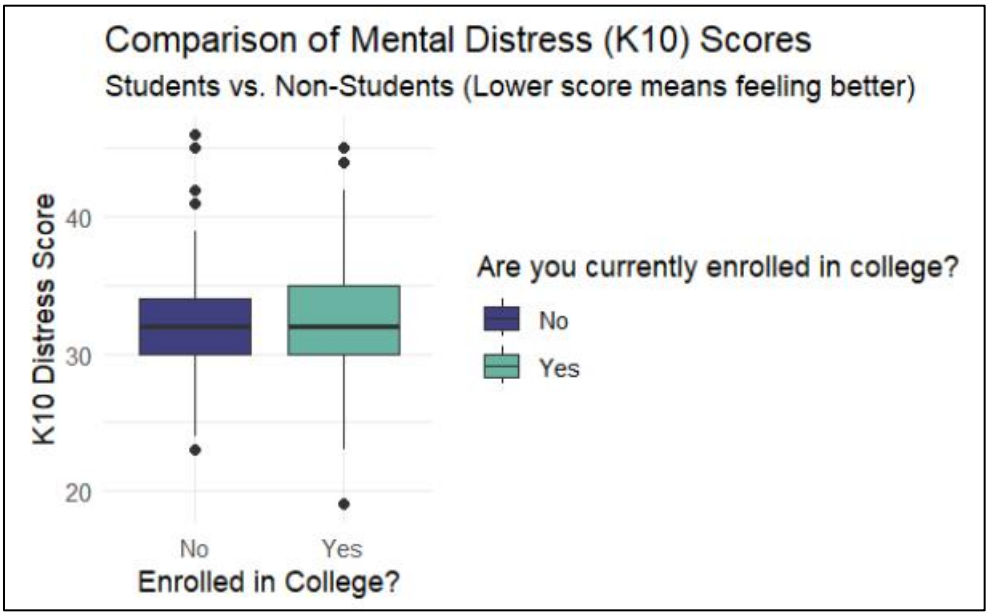

Stress Mindset Distribution by Gender
Dashed lines show mindset boundaries: Debilitating (0-10), Neutral (11-21), Enhancing (22-32)
Female
Male
Density
0.06
0.04
0.02
0.00
0
5
10
15
20
25
30
Stress Mindset Measure (SMM) Score


SMM scores by gender
SMM
30
20
10
0
Female
Male
Gender
What is your gender?
Female
Male

4.3 Q3: Comparison of Psychological Distress and Resilience by University Enrollment Status

To determine if university students experience different levels of well-being compared to non-students, a comparative analysis was conducted on Psychological Distress (K10) and Resilience (RS) scores. Preliminary checks using the Shapiro-Wilk test indicated that data for both groups were not normally distributed ($p < .05$), supporting the use of the non-parametric Wilcoxon Rank-Sum test.

- **Psychological Distress (K10):** The analysis revealed **a statistically significant difference** in distress levels between students and non-students ($p < .001$). As illustrated in the boxplot, students exhibited **a little higher median** distress scores compared to those not enrolled in college, indicating lower mental well-being among students.
- **Resilience (RS):** In contrast, no statistically significant difference was found in resilience scores based on enrollment status ($p = .099$). This suggests that being a university student **does not significantly impact** an individual's resilience or ability to recover from stress compared to the general population.



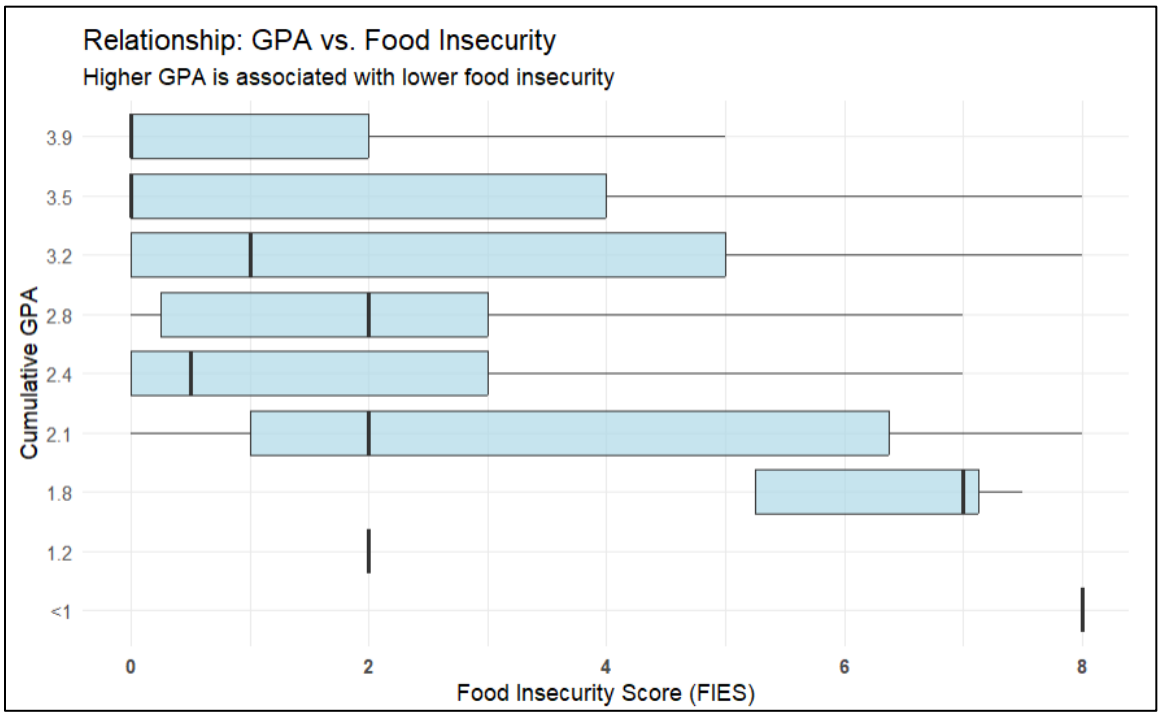
4.4 Q4: Association between Academic Performance and Food Insecurity

To investigate the connection between food availability and academic success, a linear trend analysis was conducted between students' cumulative GPA (treated as an ordered factor) and their Food Insecurity Experience Scale (FIES) scores.

Preliminary assumption checking using the Shapiro-Wilk test indicated that FIES scores were not normally distributed ($p < .001$), necessitating robust interpretation of the trends.

The linear model analysis revealed a **statistically significant negative linear trend** (Estimate (b) = -4.38, $p = .005$). The negative estimate indicates an inverse relationship: as students' GPA increases, their reported food insecurity levels significantly decrease. In other words, students with higher academic performance consistently reported better access to food compared to those with lower GPAs.

Higher-order trends, such as quadratic or cubic relationships, were not statistically significant ($p > .05$), suggesting that the improvement in food security associated with higher grades follows a linear pattern rather than a complex curve. This inverse relationship is clearly visualized in the figure down, which displays the downward trend in FIES scores across ascending GPA categories.



4.5 Q5: Impact of Household Income on Dietary Habits

To determine whether socioeconomic status influences nutritional choices, the relationship between household income levels and dietary habits (measured by Healthy and Unhealthy eating scores) was analyzed using linear model framework. While Shapiro-Wilk tests indicated non-normality for both scores ($p < .01$), the large sample size allowed for trend detection.

- **Healthy Eating:** The analysis revealed a **highly significant positive linear trend** ($b = 2.81$, $p < .001$) between income and healthy eating. As shown in the figure down, there is a clear upward trajectory: as household income increases, the consumption of nutrient-dense foods (fruits, vegetables, proteins) rises significantly. A secondary quadratic trend ($p = .015$) was also observed, suggesting that this increase may vary in intensity across income brackets but remains fundamentally positive.
- **Unhealthy Eating:** In contrast, the analysis found **no significant linear relationship** between income and unhealthy eating scores ($b = -0.08$, $p = .781$). This indicates that higher income does not simply predict higher or lower fast-food consumption. However, a significant cubic trend was detected ($p = .012$), suggesting a complex, non-linear pattern where unhealthy food consumption fluctuates—likely peaking at middle-income levels and declining at both lower and higher income extremes.